

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 29-08-2026

1. TITLE

Design Static Webpage using HTML Components

2. AIM

To design and develop a structured static webpage using HTML5 semantic elements and HTML Components.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 document.
 - To create a static webpage using HTML components.
 - To use semantic HTML elements such as <header>, <nav>, <main>, <section>, <article>, <aside> and <footer>.
 - To create navigation links within a webpage.
 - To create a user inquiry form using HTML form controls.
 - To organize webpage information using ordered and unordered lists.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. A static webpage contains fixed content that is displayed directly by a web browser.

HTML5 provides semantic elements that clearly describe the purpose of different parts of a webpage. The <header> element is used for the introductory section, <nav> contains navigation links, <main> contains the primary content, <section> groups related content, <article> represents independent content, <aside> contains supplementary information, and <footer> represents the closing section of the webpage.

HTML also provides various components for collecting user input. Form controls such as text boxes, email fields, drop-down lists, text areas, submit buttons and reset buttons can be created using HTML. In this experiment, these components are used to design a complete static webpage without external CSS.

5. ALGORITHM / PROCEDURE

- Create a new HTML file and declare the HTML5 document type using `<!DOCTYPE html>`.
- Create the `<html>`, `<head>` and `<body>` sections and set the page title.
- Create a `<header>` section displaying the portal name and welcome message.
- Create a `<nav>` section with links to the About, Students, Registration and Media sections.
- Create the About section using `<section>` and `<article>`, describing the college with bold, italic and underlined text.
- Add an ordered list `<ol>` to display the list of departments.
- Add an unordered list `<ul>` to display the list of facilities.
- Add hyperlinks under Useful Links to the college website and Google.
- Create a Students section containing a `<table>` with Roll No, Name, Branch and Email columns.
- Insert the college logo and college building images using `<img>` tags.
- Create a Student Registration Form using `<form>` and `<fieldset>` with a `<legend>`.
- Add text, email, password, radio, checkbox, select, textarea, color, date, number and range input controls to the form.
- Add Submit and Reset buttons to the form.
- Create a Media section with an `<audio>` element and a `<video>` element with playback controls.
- Create a `<footer>` containing the copyright information.
- Save the HTML file and open it in a web browser.
- Verify the webpage structure, navigation links, table, form controls and media elements.
- Compare the actual output with the expected output.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <center>
    <!--Header Section-->
    <header>
      <h1>College Information Portal</h1>
      <p>Welcome to Anits College Portal</p>
    </header>
    <hr>
    <!--Navigation Bar-->
    <nav>
      <a href="#about">About</a> |
      <a href="#students">Students</a> |
      <a href="#reg">Registration</a> |
      <a href="#media">Media</a>
    </nav>
    <hr>
    <!--Part A-->
    <section id="about">
      <article>
        <h1>Welcome to Anits</h1><br>
        <p>Building future engineers with innovation and excellence.</p><br>
        <p>
          <b>Our institute provides quality education with experienced faculty.</b><br>

```

Honoured</i>

<i>Recognized by AICTE and affiliated with a reputed university. NAAC "A+"
<u>Modern laboratories, smart classrooms, and excellent placements.</u>


```
</p>
<h1>Departments</h1>
<ol>
  <li>Computer Science</li>
  <li>Artificial Intelligence</li>
  <li>Data Science</li>
  <li>Information Technology</li>
  <li>Electronics & Communication</li>
  <li>Electrical Engineering</li>
  <li>Mechanical Engineering</li>
  <li>Civil Engineering</li>
</ol>
<h1>Facilities</h1>
<ul>
  <li>Central Library</li>
  <li>Computer Labs</li>
  <li>Sports Complex</li>
  <li>Hostel Facility</li>
  <li>Cafeteria</li>
</ul>
<h1>Useful Links</h1>
<p>College Website</p>
<a href="https://anits.org/" target="_blank">Anits Website</a>
<p>Google Search</p>
<a href="https://www.google.com" target="_blank">Google</a>

</article>
</section>
<hr>
<!--Part B-->
<section id="students">
  <article>
    <h1>Student Details</h1>
    <table border="5" cellpadding="10">
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Email</th>
      </tr>
      <tr>
        <td>201</td>
        <td>Poojith</td>
        <td>CSE</td>
        <td>poojith4521@gmail.com</td>
      </tr>
      <tr>
        <td>091</td>
        <td>Suhas</td>
        <td>AI & ML</td>
        <td>suhas45@gmail.com</td>
      </tr>
      <tr>
        <td>203</td>
        <td>Sasank</td>
        <td>IT</td>
        <td>ksasank12@gmail.com</td>
      </tr>
    </table>
  </article>
</section>
<hr>
<section>
  
```

```

        
    </section>
    <hr>
    <!--Part C-->
    <section id="reg">
        <article>
            <form>
                <fieldset>
                    <legend>
                        <h1>Student Registration Form</h1>
                    </legend>
                    <p>
                        <label for="name">Name</label>
                        <input type="text" id="name">
                        <br><br>
                        <label for="email">Email</label>
                        <input type="email" id="email">
                        <br><br>
                        <label for="password">Password</label>
                        <input type="password" id="password">
                        <br><br>
                        <label>Gender</label>
                        <input type="radio" name="gender">Male
                        <input type="radio" name="gender">Female
                        <input type="radio" name="gender">Other
                        <br><br>
                        <label>Hobbies</label>
                        <input type="checkbox">Reading
                        <input type="checkbox">Music
                        <input type="checkbox">Sports
                        <br><br>
                        <label for="branch">Branch</label>
                        <select id="branch">
                            <option>CSE</option>
                            <option>AI & ML</option>
                            <option>Data Science</option>
                            <option>IT</option>
                            <option>ECE</option>
                            <option>EEE</option>
                        </select>
                        <br><br>
                        <label for="comments">Comments</label>
                        <textarea id="comments" rows="6" cols="45" placeholder="Enter your
comments"></textarea>
                        <br><br>
                        <label for="color">Favorite Color</label>
                        <input type="color" id="color">
                        <br><br>
                        <label for="date">Date of Birth</label>
                        <input type="date" id="date">
                        <br><br>
                        <label for="number">Mobile Number</label>
                        <input type="number" id="number">
                        <br><br>
                        <label for="range">Satisfaction Level</label>
                        <input type="range" id="range">
                        <br><br>
                        <button type="submit">Submit</button>
                        <button type="reset">Reset</button>
                    </p>
                </fieldset>
            </form>
        </article>
    </section>
    <hr>
    <!--Part D-->
    <section id="media">
        <article>
            <h1>College Audio File</h1>

```

```

        <audio controls>
            <source src="college_audio.mp3" type="audio/mpeg">
        </audio>
        <h1>College Video</h1>
        <video width="220" controls>
            <source src="college_video.mp4" type="video/mp4">
        </video>
    </article>
</section>
<hr>
<footer>
    <p>&copy; 2026 Anits College Info Portal. All Rights Reserved.</p>
</footer>
</center>
</body>
</html>

```

7. CODE EXPLANATION

- <!DOCTYPE html>** – Declares that the document uses HTML5.
- <html>** – Defines the root element of the HTML document.
- <head>** – Contains metadata, viewport settings and the title of the webpage.
- <body>** – Contains the visible content of the webpage.
- <center>** – Centers the entire page content horizontally.
- <header>** – Defines the header section showing the portal name and welcome message.
- <nav>** – Contains navigation links to About, Students, Registration and Media sections.
- <section>** – Groups related content into logical sections (about, students, reg, media).
- <article>** – Represents an independent block of content within a section.
- , <i>, <u>** – Apply bold, italic and underline formatting to text describing the college.
- ** – Creates an ordered list of departments.
- ** – Creates an unordered list of facilities.
- <a href target="_blank">** – Creates hyperlinks to the college website and Google that open in a new tab.
- <table>, <tr>, <th>, <td>** – Creates the Student Details table with rows and columns.
- ** – Displays the college logo and college building images.
- <form>** – Creates the Student Registration Form for collecting user information.
- <fieldset>, <legend>** – Groups the registration form controls with a caption.
- <input type="text/email/password">** – Creates text, email and password input fields.
- <input type="radio">** – Creates Gender selection options (Male, Female, Other).
- <input type="checkbox">** – Creates Hobbies selection options (Reading, Music, Sports).
- <select>** – Creates a drop-down list to choose the Branch.
- <textarea>** – Creates a multi-line text input area for Comments.
- <input type="color">** – Creates a color picker for Favorite Color.
- <input type="date">** – Creates a date picker for Date of Birth.
- <input type="number">** – Creates a numeric input field for Mobile Number.
- <input type="range">** – Creates a slider control for Satisfaction Level.
- <button type="submit/reset">** – Creates Submit and Reset buttons for the form.
- <audio>** – Embeds the College Audio File with playback controls.

<video> – Embeds the College Video with playback controls.

<footer> – Defines the footer section containing copyright information.

8. TEST CASES AND EXECUTION DATA

The following test cases were executed on the College Information Portal and the results were recorded.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Webpage should be displayed correctly	Webpage displayed correctly	Pass
TC-02	Click About, Students, Registration and Media navigation links	Corresponding sections should be displayed	Corresponding sections displayed	Pass
TC-03	Check the Student Details table	Roll No, Name, Branch and Email should be shown for all students	All student records displayed correctly	Pass
TC-04	Enter details in the Student Registration Form (name, email, password, gender, hobbies, branch, comments, DOB, mobile, satisfaction level)	Form should accept all entered information	Information accepted	Pass
TC-05	Click the Reset button	All form fields should be cleared	Form fields cleared	Pass
TC-06	Play the College Audio and Video files	Audio and video should play with controls working	Audio and video played correctly	Pass

9. OUTPUT

College Information Portal

Welcome to Anits College Portal

[About](#) | [Students](#) | [Registration](#) | [Media](#)

Welcome to Anits

Building future engineers with innovation and excellence.

Our institute provides quality education with experienced faculty.

Recognized by AICTE and affiliated with a reputed university. NAAC "A+" Honoured
Modern laboratories, smart classrooms, and excellent placements.

Departments

1.

2.

3.

4.

5.

6.

7.

8.
- Computer Science

Artificial Intelligence

Data Science

Information Technology

Electronics & Communication

Electrical Engineering

Mechanical Engineering

Civil Engineering

Facilities

- •

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•

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- Central Library

Computer Labs

Sports Complex

Hostel Facility

Cafeteria

Useful Links

- College Website

[Anits Website](#)

Google Search

[Google](#)

Student Details

Roll No	Name	Branch	Email
201	Poojith	CSE	poojith4521@gmail.com
091	Suhas	AI & ML	suhas45@gmail.com
203	Sasank	IT	ksasank12@gmail.com

College Logo

College Building

Student Registration Form

Name

Email

Password

Gender ☐Male ☐Female ☐Other

Hobbies ☐Reading ☐Music ☐Sports

Branch

Enter your comments

Comments

Favorite Color

Date of Birth

Mobile Number

Satisfaction Level

College Audio File

College Video

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10. RESULT

Thus, a College Information Portal was successfully designed and developed as a structured static webpage using HTML5 semantic elements and HTML components. The webpage was verified for its navigation links, student details table, images, student registration form controls, and audio/video media elements, and all test cases passed successfully.